

Activity

Group members:

Diamonds data

In this class activity, we will look at data from AwesomeGems.com on the price and characteristics of 351 diamonds.

In alphabetical order, the levels of clarity in the `Diamonds` data are IF (internally flawless), SI1, SI2, SI3 (slightly included), VS1, VS2 (very slightly included), VVS1, VVS2 (very very slightly included). To be consistent with R, we will use this order for our regression model, with IF the baseline and indicator variables `IsSI1`, `IsSI2`, `IsSI3`, `IsVS1`, `IsVS2`, `IsVVS1`, and `IsVVS2`. Our model is

$$\text{Price per carat} = \beta_0 + \beta_1 \text{IsSI1} + \beta_2 \text{IsSI2} + \beta_3 \text{IsSI3} + \beta_4 \text{IsVS1} \quad (1)$$

$$+ \beta_5 \text{IsVS2} + \beta_6 \text{IsVVS1} + \beta_7 \text{IsVVS2} + \varepsilon \quad (2)$$

ANOVA table

We can use an ANOVA table to summarize the between-group and within-group variance for the `Diamonds` data. The ANOVA table below has `MSModel` and `MSE` filled in:

Source	df	SS	MS	F
Model			15.49	
Residual			8.24	
Total				

1. Complete the ANOVA table (remember that there are 351 diamonds in the data).